BT-204 (GS)

B.Tech., I & II Semester Examination, June 2025

Grading System (GS)

BASIC CIVIL ENGINEERING & MECHANICS

Time : Three Hours Maximum Marks : 70

Note : (i) Attempt any five questions. (ii) All questions carry equal marks. (iii) Original paper bilingual (Hindi + English) tha; yahan sirf English version diya gaya hai.

Coverage: Partial — Q1–Q6 mile (Q7–Q8 source me capture nahi hua)

1. (a) State the D'Alembert's principle and its application. **7**

(b) What are different characteristics of stones and their methods of testing? Write the name of various types of stones which are used in building work. **7**

2. (a) With neat sketch explain the principle of positioning with GPS. **7**

(b) Discuss the general principles that need to be observed in brick masonry construction. **7**

3. (a) Explain various types of stone masonry. Draw typical sketches to illustrate them. **7**

(b) What is mortar? Describe the estimation of mortar requirement. **7**

4. (a) Discuss the application of remote sensing in: (i) Terrain analysis (ii) Construction material inventories (iii) Site investigation. **7**

(b) State the assumptions used in simple bending and derive the bending equation. **7**

5. (a) Describe various types of windows. **7**

(b) An effort of 180 N is required just to move a certain body up an inclined plane of angle 15°, the force being parallel to the plane. If the angle of inclination of the plane is made 20°, the effort required, again applied parallel to the plane, is found to be 210 N. Find the weight of the body and co-efficient of friction. **7**

6. (a) Determine the diameter of a rod 200 m long held vertically and subjected to an axial pull of 325 kN at its lower end, if its weight per cubic metre is 80 kN and working stress is 75 MPa. Also determine the total elongation of rod. Take E = 210 GPa. **7**

(b) State work energy principle. A uniform cylinder of 125 mm radius has a mass of 0.15 kg; this cylinder rolls along the horizontal surface without slipping with a translation velocity of 20 cm/s. Determine its total kinetic energy. **7**

7. (source me ye question available nahi tha — original paper dekhein) **—**

8. (source me ye question available nahi tha — original paper dekhein) **—**

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